Recall, though, that besides cosmology's overall relevance to the story of space and time, Chapters 6 and 7 launched us into a study of the universe's early history with a specific goal: to find the origin of time's arrow. Remember from those chapters that the only convincing framework we found for explaining time's arrow was that the early universe had extremely high order, that is, extremely low entropy, which set the stage for a future in which entropy got ever larger. Just as the pages of *War and Peace* wouldn't have had the capacity to get increasingly jumbled if they had not been nice and ordered at some point, so too the universe wouldn't have had the capacity to get increasingly disordered—milk spilling, eggs breaking, people aging—unless it had been in a highly ordered configuration early on. The puzzle we encountered is to explain how this high-order, low-entropy starting point came to be.

Inflationary cosmology offers substantial progress, but let me first remind you more precisely of the puzzle, in case any of the relevant details have slipped your mind.

There is strong evidence and little doubt that, early in the history of the universe, matter was spread uniformly throughout space. Ordinarily, this would be characterized as a high-entropy configuration—like the carbon dioxide molecules from a bottle of Coke being spread uniformly throughout a room—and hence would be so commonplace that it would hardly require an explanation. But when gravity matters, as it does when considering the entire universe, a uniform distribution of matter is a rare, low-entropy, highly ordered configuration, because gravity drives matter to form clumps. Similarly, a smooth and uniform spatial curvature also has very low entropy; it is highly ordered compared with a wildly bumpy, nonuniform spatial curvature. (Just as there are many ways for the pages of *War and Peace* to be disordered but only one way for them to be ordered, so there are *many* ways for space to have a disordered, nonuniform shape, but very few ways in which it can be fully ordered, smooth, and uniform.) So we are left to puzzle: Why did the early universe have a low-entropy (highly ordered) uniform distribution of matter instead of a high-entropy (highly disordered) clumpy distribution of matter such as a diverse population of black holes? And why was the curvature of space smooth, ordered, and uniform to extremely high